OIL INDIA LIMITED - WELL COMPLETION REPORT (WCR EXTRACT)
DRILLING ENGINEERING DEPARTMENT - MORAN ASSET
SYNTHETIC TECHNICAL CASE STUDY (SIH26121 PROTOTYPE)

WELL SUMMARY:
Well Name: Moran-92 | UWI: OIL-MOR-92 | Field: Moran
Total Depth Reached: 3,920 m MD (3,845 m TVD) | Rig: Moran Super-Rig 02
Spud Date: 14-Aug-2023 | Completion Date: 02-Dec-2023

POST-WELL DRILLING HAZARD CASE STUDY: DIFFERENTIAL PIPE STICKING
1. Incident Occurrence:
At depth 2,890 m MD (Tipam Sandstone / Barail boundary), drillstring was left static for 50 minutes to conduct an MWD directional survey and gyroscopic check.
Upon attempting to resume rotation and reciprocation, string was found differentially stuck against a permeable, depleted Tipam sandstone bed.
Overbalance pressure across the sand was estimated at 620 psi (Mud weight: 10.7 ppg vs pore pressure 9.5 ppg).

2. Remedial Action & Pipe-Freeing Protocol:
- Initial Jarring: Fired hydraulic drilling jar upward with 80,000 lbs overpull. String remained stuck.
- Spotting Freeing Pill: Mixed 45 bbl of oil-based pipe-freeing soak pill containing surfactant wetting agents.
- Displaced pill to bit and allowed 3 hours soaking time across the stuck interval (2,870 m - 2,900 m).
- Re-applied upward jarring with 110,000 lbs overpull.
- After 4.5 hours of jarring and chemical soaking, drillstring freed with sudden drop in drag.
- Total NPT: 16.5 Hours.

3. Engineering Lessons & Recommendations for Offset Wells:
- Limit stationary time in open-hole depleted sands to maximum 15 minutes.
- Rotate drillpipe at 15-20 RPM during directional surveys if MWD tool permits.
- Optimize mud cake lubricity with 2-3% liquid glycol or lubricant additive to lower friction coefficient.